

Target Analysis Console

Prioritize follow-up candidates, inspect diagnostics, and move directly into the science bench.

PRIORITY QUEUE

Temperate nearby targets

OPEN THE TARGET BOARD

INSTRUMENT DECK

Climate, class, signal channels

REVIEW DIAGNOSTICS

ANALYSIS BENCH

Derived physics and plots

RUN SELECTED TARGET

Open targets

Review instruments

Run lab



habitable-zone scan

geometry check

follow-up path

WORKSPACE MODE

Choose a lab skin

ARCHIVE LINK	Connected
TARGET CARDS	Prioritized
MODEL LAYER	Ready
PUBLICATION VIEW	Available

ARCHIVE LINK

NASA archive nasa-tap catalog from NASA Exoplanet Archive TAP service is connected. Target dossiers and lab calculations are ready.

[Open JSON data](#)

PLANETS LOADED

Loaded

DISCOVERY METHODS

Varied

HZ-EVALUABLE WORLDS

Mapped

DENSITY AVAILABLE

Ready

confirmed and featured records

represented in the loaded catalog

with enough stellar/orbital information

mass and radius both known or estimated

TARGET BOARD

Prioritize candidate worlds

Start from featured systems, filter by science case, and open any target dossier for measured and modeled readouts.

Reset filters

Search planet or host

TRAPPIST, Kepler, Proxima...

Discovery method

All methods

Size class

All sizes

Science filter

All science cases

Discovery era

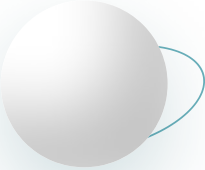
All years

Sort by

Featured worlds

Candidate targets

Click a target to analyze it in the lab.



Proxima Cen b

Around Proxima Cen

A compact world in the temperate conversation.

Climate

inside main temperate band

Body type

rocky-compatible

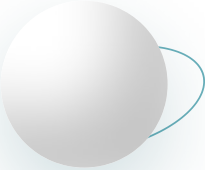
Signal

clear transit

Follow-up

measurable wobble

Conservative HZ



Barnard b

Around Barnard's star

A catalog world ready for closer inspection.

Climate

highly irradiated

Body type

volatile-rich candidate

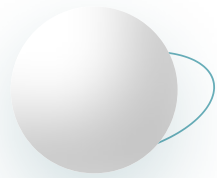
Signal

clear transit

Follow-up

delicate wobble

Terrestrial



Teegarden's Star b

Optimistic HZ

Around Teegarden's Star

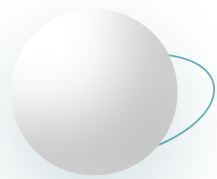
A compact world in the temperate conversation.

Climate
**near the
temperate edge**

Body type
**rocky-
compatible**

Signal
**very strong
transit**

Follow-up
**measurable
wobble**



Teegarden's Star c

Conservative HZ

Around Teegarden's Star

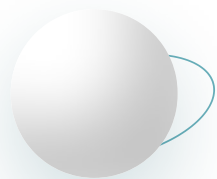
A compact world in the temperate conversation.

Climate
**inside main
temperate band**

Body type
**rocky-
compatible**

Signal
**very strong
transit**

Follow-up
**measurable
wobble**



TRAPPIST-1 e

Conservative HZ

Around TRAPPIST-1

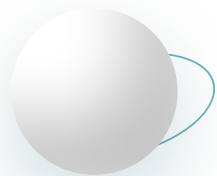
A compact world in the temperate conversation.

Climate
**inside main
temperate band**

Body type
**rocky-
compatible**

Signal
**very strong
transit**

Follow-up
**measurable
wobble**



TRAPPIST-1 f

Conservative HZ

Around TRAPPIST-1

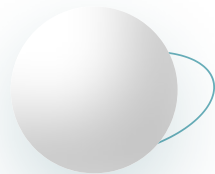
A compact world in the temperate conversation.

Climate
**inside main
temperate band**

Body type
**rocky-
compatible**

Signal
**very strong
transit**

Follow-up
**measurable
wobble**



Kepler-186 f

Conservative HZ

Around Kepler-186

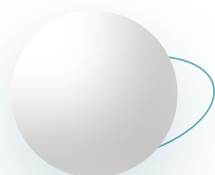
A compact world in the temperate conversation.

Climate
**inside main
temperate band**

Body type
**rocky-
compatible**

Signal
subtle transit

Follow-up
delicate wobble



TOI-700 d

Conservative HZ

Around TOI-700

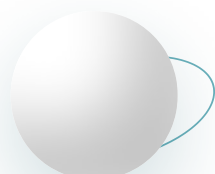
A compact world in the temperate conversation.

Climate
**inside main
temperate band**

Body type
**rocky-
compatible**

Signal
clear transit

Follow-up
delicate wobble



K2-18 b

Optimistic HZ

Around K2-18

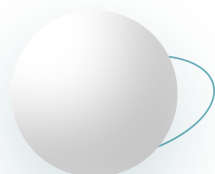
A catalog world ready for closer inspection.

Climate
**near the
temperate edge**

Body type
**volatile-rich
candidate**

Signal
clear transit

Follow-up
**measurable
wobble**



LHS 1140 b

Conservative HZ

Around LHS 1140

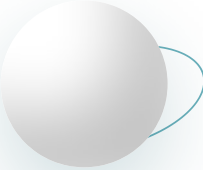
A dense world with useful mass-radius context.

Climate
**inside main
temperate band**

Body type
**rocky-
compatible**

Signal
**very strong
transit**

Follow-up
**measurable
wobble**



GJ 486 b

Around GJ 486

Super-Earth

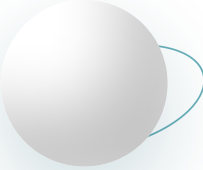
A dense world with useful mass-radius context.

Climate
highly irradiated

Body type
rocky-compatible

Signal
clear transit

Follow-up
measurable wobble



GJ 1214 b

Around GJ 1214

Sub-Neptune

A strong candidate for atmosphere-focused follow-up.

Climate
highly irradiated

Body type
volatile-rich candidate

Signal
very strong transit

Follow-up
measurable wobble

Load more

TARGET DOSSIER

TRAPPIST-1 e

TRAPPIST-1 · Terrestrial · Transit

Method	Transit
Discovery era	space-survey wave
Size family	Terrestrial
System distance	nearby system
Host star	cool red host
Facility	Multiple Observatories
Signal readout	
Climate read	inside main temperate band
Bulk character	rocky-compatible
Transit signal	very strong transit

RV follow-up	measurable wobble
Atmosphere signal	possible spectrum
Composition	Rocky-compatible bulk density

No curated description is available for this record. Refreshing from NASA TAP preserves core physical and discovery fields.

Open in lab

INSTRUMENT DECK

Select an analysis instrument

The first layer behaves like a science control room: pick the diagnostic you need, then open a target for the exact measurements and derived model readouts.

Climate window

thermal triage

Separate high-irradiation records from candidates worth climate inspection.

Mass-radius regime

population plot

Inspect the loaded catalog against density contours and structural regimes.

Python/Matplotlib SVG generated from the loaded JSON catalog.

[Open full-size SVG](#)

Detection channel

survey methods

Understand whether the planet entered the archive through shadow, wobble, lensing, or direct light.

Python/Matplotlib SVG generated from the loaded JSON catalog.

[Open full-size SVG](#)

Target field

log-log field

Scan radius against incident stellar flux with a labelled temperate-flux interval.



Archive cadence

survey wave

Track how the target pool shifts from early ground campaigns to current survey records.



INTERACTIVE ANALYSIS BENCH

Selected-target science lab

The science figures are generated by Python/Matplotlib during the data-refresh workflow, then served as responsive SVGs that use the full browser width where possible. Change albedo and atmospheric mean molecular weight to update the numerical readout; the plotted observing diagnostics use catalog parameters.



Planet

Bond albedo: 0.30

Atmosphere model

Transmission scale heights: 5

TRAPPIST-1 e — TRAPPIST-1

Earth-like N₂/O₂, $\mu \approx 29$

TARGET

TRAPPIST-1 e

TRAPPIST-1 · Transit · 2017

Inside conservative HZ

Incident flux

0.646 S $\oplus$

Luminosity divided by orbit squared

Model T_{eq}

228.4 K

albedo 0.30

Bulk density

4.9 g/cm³

Rocky-compatible bulk density

Surface gravity

0.818 g $\oplus$

vesc 9.7 km/s

Transit depth

5,007 ppm

central duration $\approx$ 0.946 h

RV semi-amplitude

1.208 m/s

edge-on unless inclination is known

Transmission proxy

69.7 ppm

H $\approx$ 8.2 km

Astrometric wobble

0.054 μ s

maximum angular reflex signal

Stellar luminosity

5.53e-4 L $\odot$

R $\star$ 0.119 R $\odot$

Semi-major axis

0.0293 AU

P 6.101 d

Rocky CMF proxy

Not applicable

valid mainly for 1–8 M $\oplus$ rocky planets

Archive T_{eq}

249.7 K

literature/archive field

Habitable-zone placement

orbit vs. climate boundaries

Python/Matplotlib SVG generated during the data-refresh workflow.

[Open full-size SVG](#)

Transit light curve

limb-darkened geometry

Python/Matplotlib SVG generated during the data-refresh workflow.

[Open full-size SVG](#)

Radial velocity curve

Keplerian model

Python/Matplotlib SVG generated during the data-refresh workflow.

[Open full-size SVG](#)

Mass-radius context

density contours

Python/Matplotlib SVG generated during the data-refresh workflow.

[Open full-size SVG](#)

Assumptions and caveats

- Kopparapu HZ coefficients are formally calibrated for roughly 2600–7200 K stars; this target falls outside or lacks Teff.
- Transit duration assumes a circular, central transit and ignores limb darkening.
- RV amplitude uses a simplified Keplerian semi-amplitude and assumes $\sin(i)=1$ when inclination is unknown.
- Transmission signal is a scale-height proxy, not an atmospheric retrieval or detectability guarantee.
- The lab figures prefer static Python/Matplotlib SVGs generated during data refresh; browser SVG fallbacks are used only when a target-specific plot pack is unavailable.

EXPLAINABLE ASTRONOMY

A compact exoplanet physics chapter

The interface now treats the model layer like a textbook page: display equations, symbols, units, and assumptions are separated from the prose. These are simplified observing equations for public science and target triage, not a replacement for a full retrieval or orbital fit.

★ stellar quantity **p** planetary quantity ⊕ Earth-normalized unit ☉ solar-normalized unit

01 Incident stellar flux

How strongly the planet is irradiated relative to Earth.

$$S_p = \frac{L_\star / L_\odot}{(a/\text{AU})^2} \quad (1)$$

A planet with $S_p = 1$ receives the same bolometric flux Earth receives from the Sun. This is the quantity used for the radius-flux population map and for a first-order climate triage.

$L_\star$ stellar luminosity

a semi-major axis

S_p planet incident flux in Earth units

02 Equilibrium temperature

A zero-dimensional temperature estimate controlled by albedo.

$$T_{\text{eq}} = T_{\text{eff}} \left(\frac{R_\star}{2a} \right)^{1/2} (1 - A_B)^{1/4} \quad (2)$$

The albedo slider changes A_B . The expression assumes global heat redistribution and no greenhouse correction, so it is a diagnostic temperature rather than a climate model.

T_{eff} stellar effective temperature

$R_\star$ stellar radius

A_B Bond albedo

03 Kopparapu-style habitable-zone boundaries

Climate boundaries expressed as stellar-flux limits and then converted to orbital distance.

$$S_{\text{eff}} = S_{\text{eff},\odot} + aT_\star' + bT_\star'^2 + cT_\star'^3 + dT_\star'^4 \quad (3)$$

The site uses inner and outer flux limits to classify a planet as inside, hotward of, or coldward of the optimistic and conservative habitable-zone bands.

$$d_{\text{HZ}} = \left(\frac{L_\star / L_\odot}{S_{\text{eff}}} \right)^{1/2} \text{AU}, \quad T_\star' = T_{\text{eff}} - 57$$

The plotted HZ panel shows the orbit against those boundaries; it does not imply habitability without atmosphere, geophysics, stellar activity, and composition constraints.

04 Transit depth and duration

Photometric observability from the ratio of planetary and stellar radii.

$$\delta_0 \simeq k^2, \quad k = \frac{R_p}{R_\star} \quad T_{14} \simeq \frac{P}{\pi} \arcsin \left[\frac{2k}{1+k^2} \right]$$

The Python light-curve figure adds a simple limb-darkened shape around this geometric depth. The catalog readout reports depth in parts per million.

k radius ratio

b impact parameter

i orbital inclination

06 Bulk density, gravity, and escape velocity

First-order structural diagnostics from mass and radius.

$$\rho_p = \rho_\oplus \frac{M_p/M_\oplus}{(R_p/R_\oplus)^3} \quad v_{\text{esc},p} = v_{\text{esc},\oplus} \left(\frac{M_p/M_\oplus}{(R_p/R_\oplus)^3} \right)^{1/2}$$

These quantities help separate likely rocky worlds, volatile-rich sub-Neptunes, and gas giants. They are most meaningful when mass and radius are both measured rather than inferred.

05 Radial-velocity semi-amplitude

The stellar reflex velocity induced by an orbiting planet.

$$K = \left(\frac{2\pi G}{P} \right)^{1/3} \frac{M_p \sin i}{(M_\star + M_p)^{2/3}} \frac{1}{\sqrt{1-e^2}}$$

When inclination is missing, the app uses an edge-on approximation. The plotted curve is a Keplerian observing model from catalog parameters, not a fit to raw spectra.

Earth-unit shortcut:

$$K \approx 0.08945 \text{ m s}^{-1} (M_p/M_\oplus)(P/\text{yr})^{-1/3} (M_\star/M_\odot)^{-2/3} \sin i / \sqrt{1-e^2}$$

A scale-height estimate for atmosphere follow-up prioritization.

$$H = \frac{k_B T_{\text{eq}}}{\mu m_H g_p} \quad (7a)$$

The atmosphere menu changes μ , the mean molecular weight. The scale-height slider changes N , the number of scale heights assumed to contribute to spectral modulation.

$$\Delta_\lambda \sim \frac{2NHR_p}{R_\star^2} \quad (7b)$$

This is intentionally a proxy: clouds, hazes, composition, stellar contamination, and instrument systematics can dominate real observations.

HOW EXOPLANETS ARE DETECTED

Discovery methods connected to observables

Each method measures a different physical signature, so the science lab reports the quantities that matter for follow-up rather than only the discovery label.

Transit photometry

Detects a small, periodic dimming when a planet crosses the stellar disk. The depth constrains radius; repeated events constrain period; duration and shape constrain orbital geometry.

Radial velocity

Measures Doppler shifts in stellar spectra caused by the star orbiting the system barycenter. It gives a mass-related signal and is strongest for massive planets close to low-mass stars.

Direct imaging

Separates faint planetary light from a bright host star. It favors young, hot, massive planets at wide separations where contrast and angular separation are less punishing.

Microlensing

Uses the gravitational field of a foreground system as a lens. It can reveal planets far away, but the events are usually non-repeatable and follow-up characterization is difficult.

OPEN DATA WORKFLOW

Built for reproducible public publishing

The repository includes a Python script that queries the NASA Exoplanet Archive TAP service, converts selected fields into a browser-ready JSON catalog, and keeps the static site deployable through GitHub Pages.

[View repository](#)

1. Static site

No build step is required. The browser loads `index.html`, `styles.css`, `app.js`, and `data/exoplanets.json`.

2. Science fields

The updater requests stellar mass, stellar radius, luminosity, inclination, eccentricity, insolation, density, and photometric fields when available.

3. Data refresh

Run `python scripts/fetch_exoplanets.py` locally or trigger the included GitHub Actions workflow to update and redeploy the JSON catalog.

**NASA Exoplanet
Archive**

**NASA TAP
documentation**

**Kopparapu HZ
calculator**

NASA Exoplanets

Built by Arif Solmaz as an open, public GitHub Pages exoplanet research lab. Use the included archive workflow for official NASA-derived values; treat browser-side estimates as educational approximations.